EC-390L Digital Signal Processing

BS (CE)-2023

PROJECT REPORT

Submitted By:

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Project Title:

Stress Monitoring via PPG and GSR

Introduction:

Stress affects both mental and physical health, and continuous monitoring is important for early detection. Physiological signals such as PPG (Photoplethysmography) and GSR (Galvanic Skin Response) are widely used indicators of stress. In this project, a software-based system is developed to simulate, analyze, and visualize PPG and GSR signals. Using basic DSP techniques, the system processes these signals to identify stress conditions and display results through graphs.

Objective:

- To use PPG and GSR signals showing normal and stressed conditions.
- To process these signals to find heart rate and changes in skin conductance.
- To detect stress by applying simple rules and signal processing techniques.
- To show and compare the original and processed signals in graphs.
- To build a small demo system that monitors stress using simulated data.

Procedure:

- Download PPG and GSR signals from online datasets (e.g., Kaggle or INRIA).
- Load the signal data into Python.
- Apply filtering on PPG to remove noise.
- Perform peak detection on the PPG signal to calculate heart rate.
- Analyze GSR signal changes to identify rapid rises in skin conductance.
- Plot raw and processed signals for comparison and highlight stress events.
- Display final output showing whether the subject is stressed or normal.

Methodology:

I collected PPG and GSR signals and loaded them using Python. The signals were preprocessed to remove noise, and important features like heart rate and GSR values were extracted. Using these features, I analyzed stress levels and classified them as Relaxed, Mild, or High. Finally, I visualized the results and generated a report showing the stress assessment.

Tools & Software:

- **Python** for data loading, preprocessing, feature extraction, and analysis
- **Pandas** for handling and managing the dataset
- **NumPy** for numerical computations

- **SciPy / Signal Processing Libraries** for filtering signals (Savitzky–Golay)
- **Matplotlib** for visualizing signals and results

DSP Techniques Used:

1. **Savitzky-Golay Filter:** Smooth PPG and GSR signals to remove noise.

$$y_{\text{smooth}} = \sum c_j \cdot y_{i+j}$$

2. **Peak Detection:** Detect heartbeats from PPG signal.

$$HR = \frac{60}{\text{average time between peaks}}$$

3. **GSR Change Detection:** Measure stress level from skin conductance variations.

$$\Delta GSR = |GSR_{i+1} - GSR_i|$$

4. **Stress Level Calculation:** Combine HR and GSR to get stress percentage.

$$Stress\% = 0.6 \cdot HR + 0.4 \cdot GSR$$

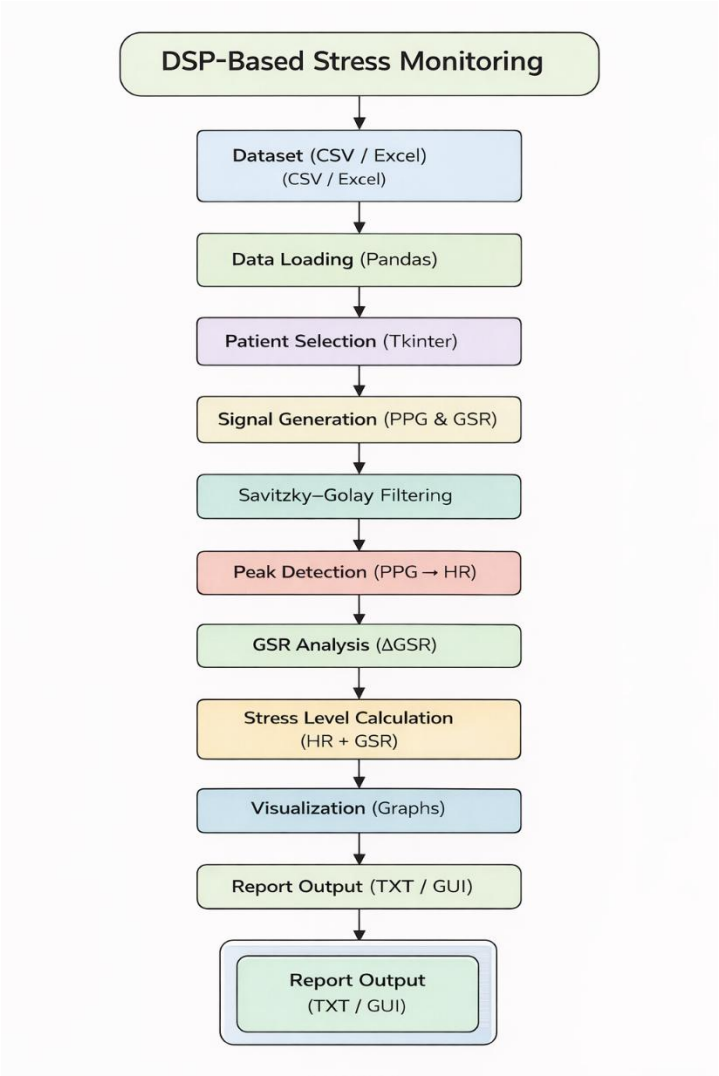
Data set:

- Signal files from kaggle with the name Stress.csv.

Limitations and Applications:

- Useful for health monitoring and early stress detection using PPG and GSR signals
- Can be applied in workplaces, education, and wearable health devices
- Provides clear visualization of stress patterns through signal processing
- System is based on simulated data, not real-time hardware implementation
- Accuracy depends on signal quality and simple threshold-based detection

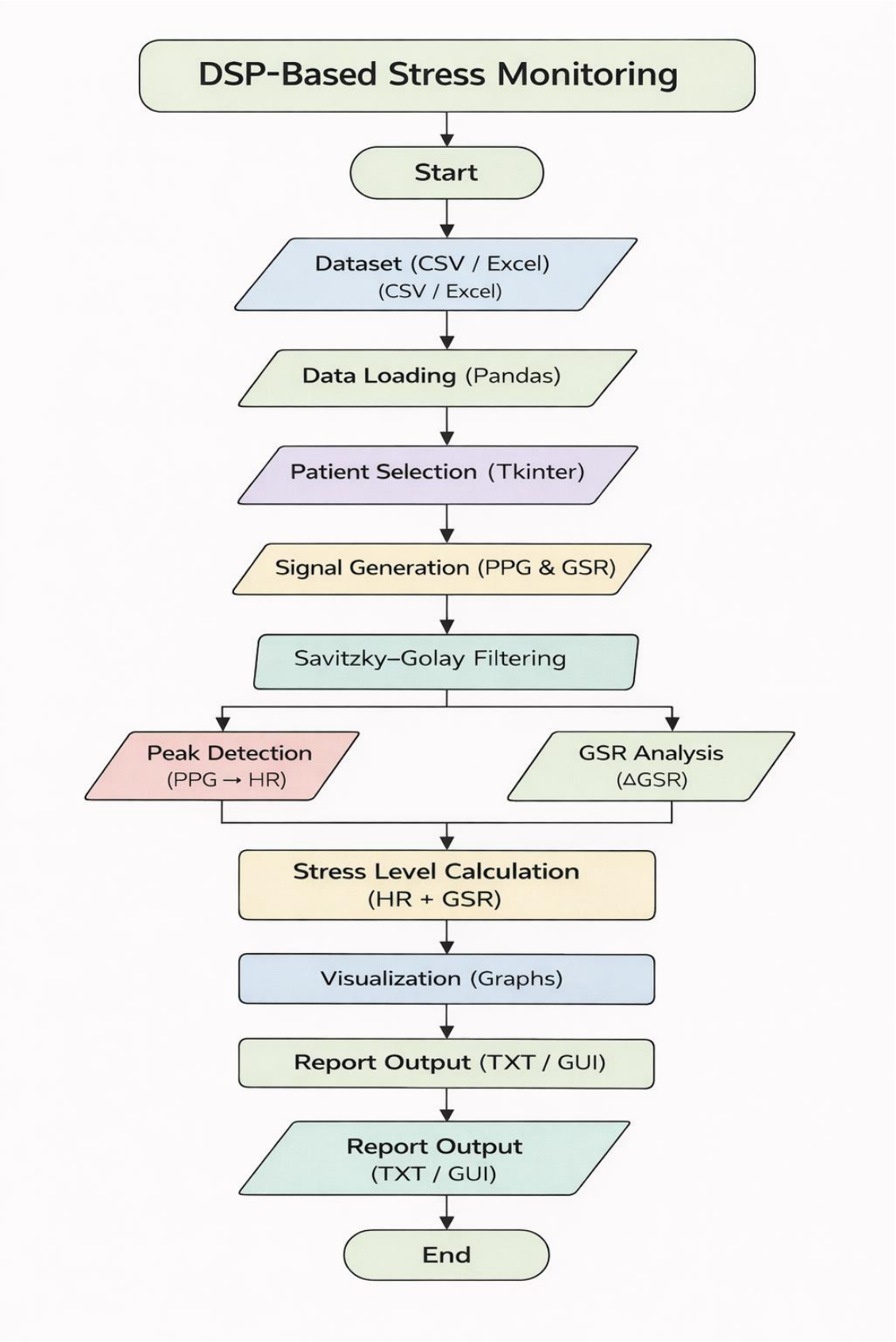
Block Diagram:



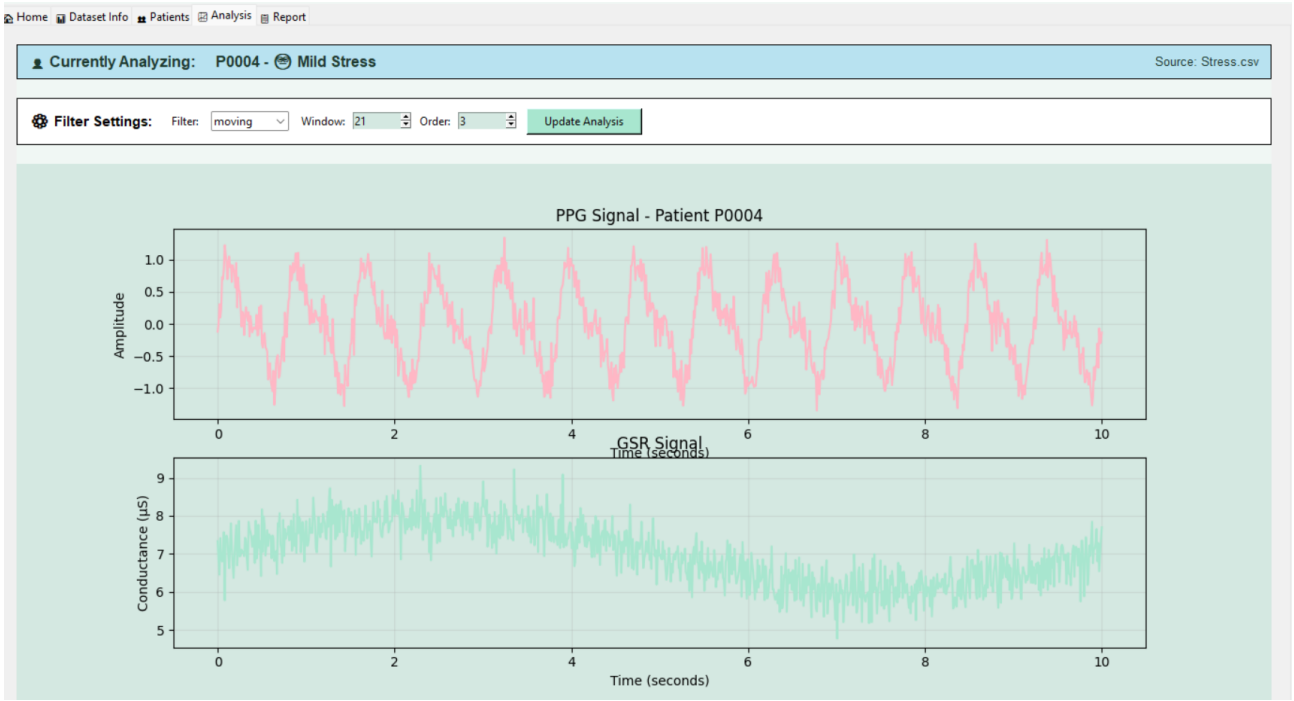
Future Work:

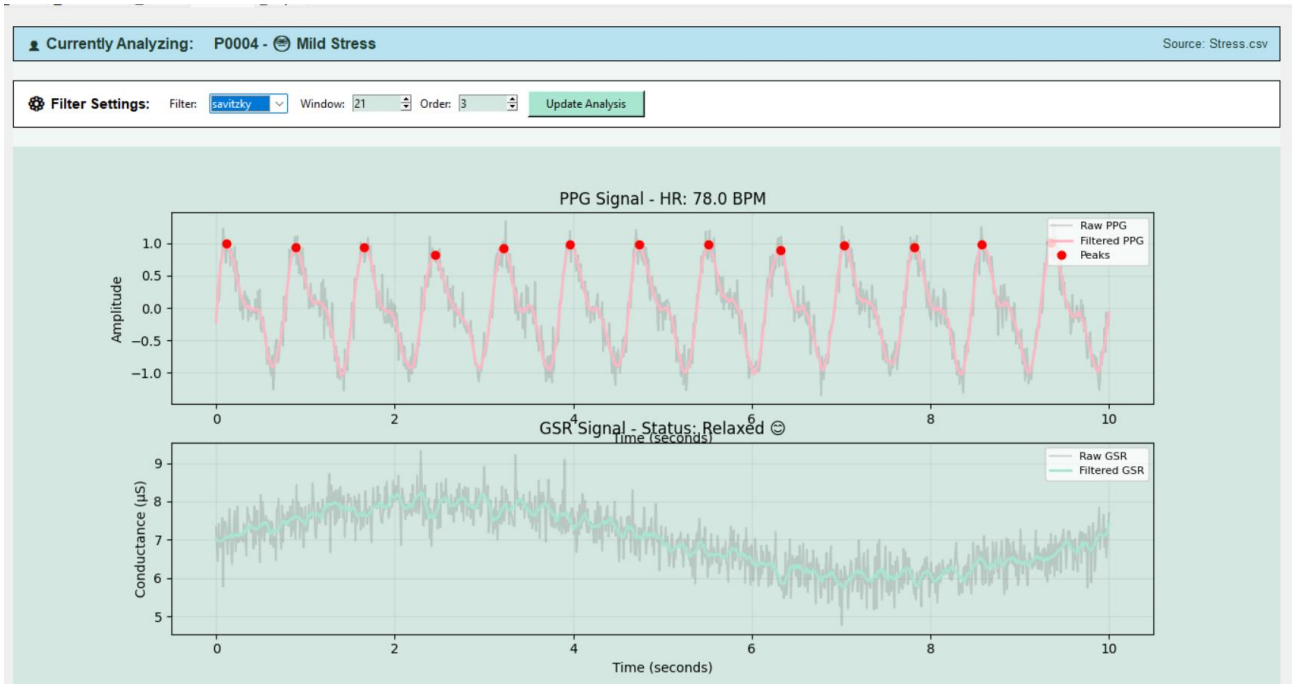
- Collect real-time PPG and GSR signals using wearable devices.
- Use advanced signal processing and machine learning for better stress detection.
- Extend the system for real-time monitoring and alerts.
- Include additional physiological signals like ECG or respiration.
- Develop a mobile or web application for easier access.

Flow Chart


DSP-Based Stress Monitoring
Start
Dataset (CSV / Excel)
(CSV / Excel)
Data Loading (Pandas)
Patient Selection (Tkinter)
Signal Generation (PPG & GSR)
Savitzky–Golay Filtering
Peak Detection
(PPG → HR)
GSR Analysis
(ΔGSR)
Stress Level Calculation
(HR + GSR)
Visualization (Graphs)
Report Output (TXT / GUI)
Report Output
(TXT / GUI)
End

Plot of Raw and Processed Signals:





Conclusion:

The DSP-based system effectively analyzes PPG and GSR signals to detect stress levels. Filtering and peak detection improve signal accuracy, providing reliable stress monitoring and easy visualization for healthcare and personal use.